

Epic / ClickUp task setup ▾

Okay I have a new thing to do on ClickUp. Help me with it

I'd be happy to help you add a task to ClickUp! What's the task you want to create?

no check clickup it should be in progress

Loaded tools, used ClickUp integration >

Here are your current **in progress** tasks:

1. **Define C++ component scope: CLI client with TLS sockets / OOP message store / CMake build**
 - Assigned to: Matthew Burke
 - List: Sprint 1: Planning & Architecture
2. **Read full EPIC brief - highlight every rubric criterion and mark requirement**
 - Unassigned
 - List: Sprint 1: Planning & Architecture

What's the new thing you want to add or do?

1

Got it — what do you want to do with task 1? Update it, add a comment, mark it complete, something else?

No I want your help actually doing the task

Sure! Looking at the task — **Define C++ component scope: CLI client with TLS sockets / OOP message store / CMake build** — you've already got a solid foundation in your codebase. Let me give you a clear scope definition based on what's there and what the rubric needs.

From what I can see in your code, you already have the skeleton for:

- `Client.hpp/.cpp` — main window + HTTP controller
- `User.hpp/.cpp` — identity + keypair (libsodium)
- `Message.hpp/.cpp` — data holder
- `MessageStore.hpp/.cpp` — crypto + cache
- `CMakeLists.txt` — Qt6 + libsodium + libcurl

The question is what to actually implement given your time before June 3rd. What would be most useful — should I help you:

1. **Prioritise which TODOs to implement** (e.g. keypair generation, encrypt/decrypt, the HTTP calls) based on what will score best in Kashif's rubric
2. **Actually write the C++ implementation** for specific parts (e.g. `User::generateKeyPairs()`, `MessageStore::encryptMessage()`)
3. **Both** — decide scope first, then start writing code

I have a good bit of time - I think we should decide scope / what we want things to do - we want a better frontend than just a CLI, so def be something better than that - also does a large part of the crypto not have to happen client side... so we will be using C++ for that? but then if our API is written in Python is that going to be a problem?

These are really good questions — let me address them:

On the frontend: Yes, avoid CLI. Your CMakeLists already pulls in Qt6 Widgets, so a proper Qt GUI is the right call. That gives you a real window, login form, chat view, contact list — and it'll look much more impressive at the demo.

On crypto and the C++ client: You're thinking about this correctly. The whole point of E2EE is that encryption/decryption happens on the *client*, so the server never sees plaintext. So yes — the C++ client should handle:

- Key generation (X25519 + Ed25519 via libsodium)
- Encrypting messages before sending to the API
- Decrypting messages received from the API
- Signing/verifying

On Python backend vs C++ client: This is absolutely not a problem. The backend is just a dumb relay — it stores ciphertext blobs and serves them back. It never needs to touch the crypto. The C++ client and Python server communicate over HTTP/JSON, which is language-agnostic. Your `Client.cpp` already uses libcurl for exactly this.

So the architecture is clean:

- **C++ client** — Qt GUI + libsodium crypto + libcurl HTTP
- **Python backend** — stores/relays ciphertext, handles auth, pushes hashes to blockchain
- **Web client (JS)** — same idea but using Web Crypto API

The main question for scoping is: do you want the C++ client to be the *primary* client, or is the web frontend the primary one and C++ is supplementary? That affects how much effort goes into each.

okay - so what parts of the project have crypto involved is it basically just in the C++ client?

No, crypto is actually spread across multiple parts. Here's the breakdown:

C++ Client (libsodium)

- Key generation — X25519 + Ed25519 keypairs
- Encrypting messages before sending
- Decrypting received messages
- Signing ciphertext / verifying signatures
- Saving/loading private keys encrypted on disk (Argon2id KDF + secretbox)

Web Client / JS Frontend (Web Crypto API)

- Same thing as above but for browser users
- `crypto.js` already has the stubs — `generateKeyPair()`, `encryptMessage()`, `decryptMessage()`, `signMessage()` etc.

- Private keys stored in IndexedDB

Python Backend

- Password hashing with Argon2id on registration — already stubbed in `auth_service.py`
- JWT signing/verification — already partially implemented
- Computing keccak256 hash of each ciphertext before pushing to blockchain
- It does **not** do message crypto — it just stores and relays ciphertext

Blockchain (Solidity)

- keccak256 hashing is done natively by the EVM
- The smart contract just stores and verifies those hashes

So really there are **two full crypto implementations** — one in C++ and one in JS — doing essentially the same thing from different clients. The Python side is much lighter, mostly just password hashing and JWT.

Given the rubric, Eoin's cryptography marks care about the *design and correctness* across the whole system, not just one client. Does that change how you're thinking about splitting work in the team?

nah its fine - so should i write this down anywhere? is it already in the read me ... whats the story - how do i complete this task and get to writing code

Your README already has a really solid encryption scheme section — X3DH, Double Ratchet, the key bundle structure, the send/receive flows. And `crypto_setup.txt` has the full crypto design. So the *crypto design* is well documented.

What's missing for this task specifically is a C++ **component scope document** — basically a clear statement of "here's exactly what our C++ client will do, what classes implement what, and what we're committing to building." That's what the task is asking you to define.

I'd suggest completing it like this:

1. **Write a short scope doc** (could just be a section added to the README, or a separate `client-cpp/README.md`) that states:
 - What the Qt GUI looks like (login view, chat view)
 - Which crypto operations the C++ client owns
 - What each class does (`User`, `Message`, `MessageStore`, `Client`)
 - What the libcurl HTTP calls do
 - What you're *not* implementing (e.g. no local SQLite persistence, no WebSocket)
2. **Mark the ClickUp task done**
3. **Start a new task for actual implementation** — probably broken into subtasks like "implement `User::generateKeyPairs()`", "implement `MessageStore::encryptMessage()`", "build Qt login UI" etc.

Want me to draft that scope doc now? I can write it as a `client-cpp/README.md` based on everything I can see in your codebase, and then we can move straight into implementation.

Yes please

Viewed a file, created a file, read a file >

There's your scope doc — drop it into `client-cpp/README.md` in your repo. It covers the GUI layout, every class and its responsibilities, all the libsodium functions mapped to where they're called, the HTTP endpoints, and a suggested implementation order.

That implementation order at the bottom is probably the most useful bit — it gives you a natural sequence to start writing code. Want to kick off with step 1,

`User::generateKeyPairs()`?



Readme
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are these the ephemeral keys in X3DH?:

generate X25519 (encryption) and Ed25519 (signing) keypairs using

No, these are the **long-term identity keys**, not ephemeral ones.

Here's the distinction:

Long-term keys (generated once at registration, stored on disk):

- `X25519` keypair → Identity Key (IK) — used in X3DH DH operations
- `Ed25519` keypair → Signing key — signs the Signed Prekey (SPK) so recipients can verify the bundle came from you

These are what `User::generateKeyPairs()` produces and what `saveKeyFile()` / `loadKeyFile()` persist encrypted to disk.

Ephemeral keys (generated fresh for each message send, never stored):

- A new X25519 keypair generated on the fly inside `MessageStore::encryptMessage()`
- Used for one DH operation then immediately zeroed with `sodium_memzero()`
- The ephemeral *public* key gets sent along with the ciphertext so the recipient can derive the same shared secret
- This is what gives you forward secrecy — even if your long-term key is later compromised, past messages can't be decrypted because the ephemeral private key is gone

So in the full X3DH picture your C++ client needs to handle:

Key	Type	Where generated	Where stored
Identity Key (IK)	Long-term X25519	<code>User::generateKeyPairs()</code>	Encrypted keyfile on disk
Signing Key	Long-term Ed25519	<code>User::generateKeyPairs()</code>	Encrypted keyfile on disk
Signed Prekey (SPK)	Medium-term X25519	Needs implementing	Uploaded to server, rotated weekly
One-Time Prekeys (OPKs)	Single-use X25519	Needs implementing	Uploaded to server in batches
Ephemeral Key (EK)	Per-message X25519	<code>MessageStore::encryptMessage()</code>	Never stored — zeroed after use

Your current scope doc actually undersells the key management a bit — the SPK and OPK generation is a meaningful chunk of work. Worth knowing before you dive in.

okay so X25519 is the IK and Ed25519 is the SPK that changes weekly?

Not quite — let me clear that up:

X25519 is the *curve* used for Diffie-Hellman key exchange. It's used for:

- Identity Key (IK) — long-term
- Signed Prekey (SPK) — medium-term, rotated weekly
- One-Time Prekeys (OPKs) — single use
- Ephemeral Key (EK) — per message

So X25519 is used for **all four** of those key types. They're all DH keys, just with different lifetimes.

Ed25519 is a completely different curve, used purely for **digital signatures**. It's not used for DH at all. Its only job is:

- Sign the SPK so recipients can verify the bundle came from the real Bob and not the server

So the relationship is:

- Alice fetches Bob's SPK (which is an X25519 key)
- Alice checks the Ed25519 signature on that SPK using Bob's Ed25519 public key
- If the signature is valid, she trusts the SPK is genuine and proceeds with X3DH

Think of Ed25519 as the "vouching" key — it doesn't participate in the key exchange itself, it just proves the X25519 keys are legitimate.

So to directly answer your question: the SPK is **X25519**, and Ed25519 is a separate signing-only key that lives alongside the IK and never rotates (in the standard Signal design).

AI_Prompt_Artefacts.docx

73 lines

DOCX

Look back through this conversation and identify every message I (the human) sent that could serve as an AI prompt artefact for a university submission. I need prompts that demonstrate: - Security-driven thinking (e.g. specifying exact crypto parameters, security properties, failure modes) - Critical judgement (e.g. constraining scope, correcting a previous output, rejecting something) - Technical specificity (e.g. naming exact files, algorithms, RFCs, line numbers) For each one you find, return: 1. The exact text of my message Skip anything that's just a short question or command like "give me the cmds" or "yes do that". I only want substantive prompts. can you add them to this file , just the straight prompts

Read the SKILL.md for docx >

Now let me copy the uploaded file and add the new prompts from this conversation:

Ran 2 commands, viewed a file >

Now I have the full picture. The qualifying prompts from this conversation are the ones about the C++ scope/crypto architecture. Let me identify them and add them:

From this conversation, the qualifying prompts are:

- "I have a good bit of time - I think we should decide scope... we want a better frontend than just a CLI... does a large part of the crypto not have to happen client side... so will we be using C++ for that? but then if our API is written in Python is that going to be a problem?"
- "okay - so what parts of the project have crypto involved is it basically just in the C++ client?"
- "are these the ephemeral keys in X3DH?: generate X25519 (encryption) and Ed25519 (signing) keypairs using"
- "okay so X25519 is the IK and Ed25519 is the SPK that changes weekly?" (this is a correction/critical judgement moment — checking and probing a technical claim)

Ran 2 commands, edited a file, read a file >

Added prompts 14-17 from this conversation:

- **14** — Scoping the C++ client: questioning whether crypto must be client-side and whether Python API is a conflict
- **15** — Probing where crypto lives across the full system (not just C++)
- **16** — Questioning whether **X25519 + Ed25519** generation refers to ephemeral X3DH keys — good example of verifying a technical claim
- **17** — Challenging the distinction between X25519 (IK) and Ed25519 (SPK) — another good critical correction moment



Ai prompt artefacts updated
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